Review - 2

VEHICLE ACCIDENT DETECTION SYSTEM

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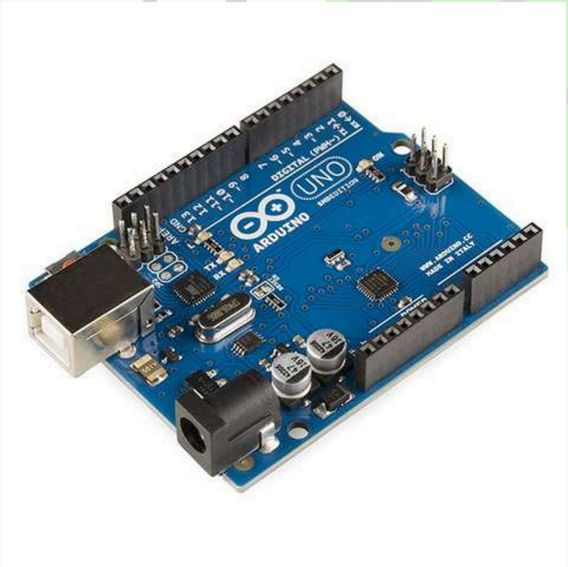
Project Objective:

1. Detect accidents automatically
2. Prevent drunk driving
3. Send GPS location during emergencies
4. Allow manual cancel for false detection

Main Components:

1. Arduino UNO
2. MQ3 Alcohol Sensor
3. ADXL Accelerometer/Gyroscope
4. GPS Module
5. GSM Module
6. I2C LCD Display
7. Relay Module
8. Buzzer
9. Push Button
10. DC Motor
11. Switch
12. Battery

Arduino UNO :



1. Arduino UNO is the main controller of the project.

2. It reads data from sensors and controls all modules like GSM, GPS, LCD, relay and buzzer
   - If alcohol detected → stop vehicle
   - If accident detected → send alert message

Alcohol Sensor (MQ3):



1. Detects alcohol level in driver's breath.
2. If alcohol exceeds threshold:

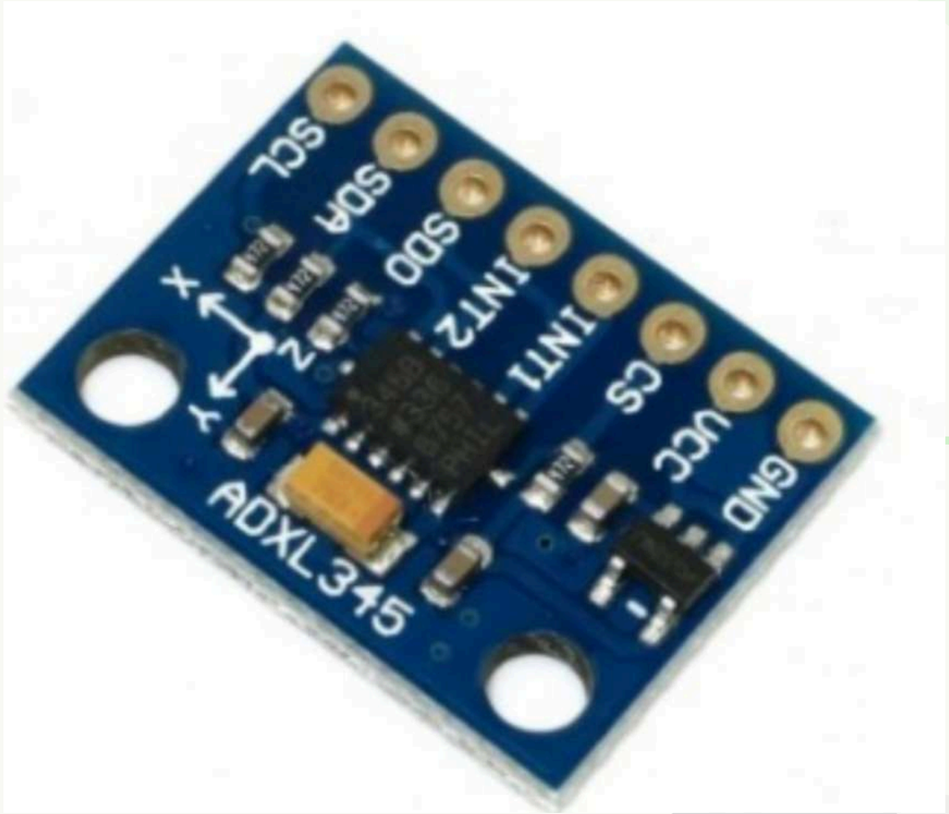
– Engine is locked using relay
– Warning alert is activated.

MQ3Pin	Arduino UNO
VCC	5V
GND	GND
AO	A0
DO	optional

Accelerometer / Gyroscope (ADXL):

1. Detects sudden acceleration and tilt.
2. Used to identify crash or accident conditions.
3. Sends motion data to Arduino.

ADXL Pin	Arduino UNO
VCC	3.3V
GND	GND
SDA	A4
SCL	A5
CS	3.3V
SD	GND



GPS Module:

1. Provides real-time location coordinates.
2. Latitude and longitude are sent during emergency

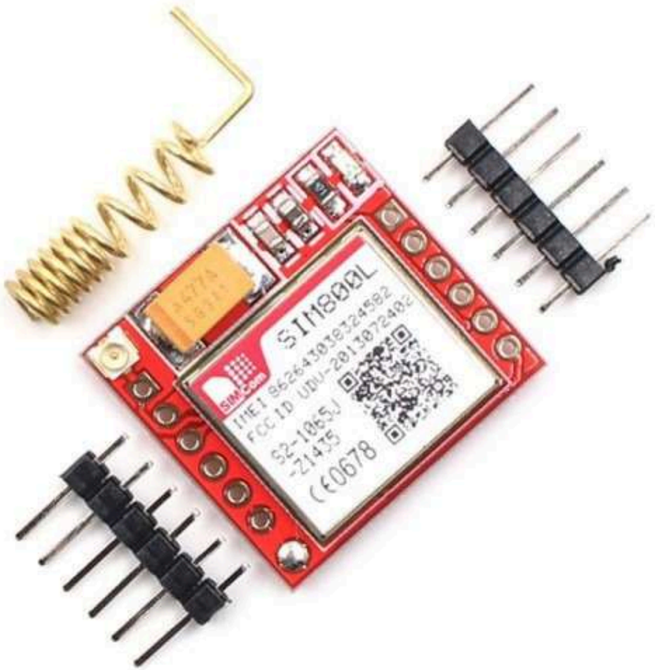


GPS Pin	Arduino UNO
VCC	5V
GND	GND
TX	D4
RX	D2

GSM Module:

1. Sends SMS or call during accidents.
2. Emergency message contains GPS location.
   - Uses SIM card to send SMS or make calls.
   - Controlled through AT commands.

GSM Pin	Arduino UNO
VCC	External 4V
GND	GND
TXD	D10
RXD	D9



LCD Display :

1. Display system status
2. Shows sensor readings and alerts and Helps users monitor system operation visually.
3. Connected using I2C (SDA, SCL).

LCD Pin	Arduino UNO
LCD	UNO
GND	GND
VCC	5V
SDA	A4
SCL	A5



Relay Module

- Arduino cannot directly power motors.
- Relay acts as an electrical switch.
- Controls high-power devices.

Function in project

- Stops the DC motor (vehicle engine simulation) when:
  - Alcohol detected
  - Accident detected



Relay	Arduino UNO
VCC	5V
GND	GND
IN	D7

Buzzer Module

- Provides audio alerts.
- Produces sound when activated by Arduino.
- Alarm when:
  - Alcohol detected
  - Accident detected

BUZZER	**Arduino UNO**
IN	D6
GND	GND



Push Button

- Manual control input.
- Allows user interaction with the system.
- Can be used to:
  - Cancel false accident alert
  - Reset system



PUSH	Arduino UNO
PIN1	D2
PIN2	GND

Motor

- Simulates vehicle movement.
- Represents the vehicle engine or motion.
- Runs normally when system is safe.
- Stops when:
  - Alcohol detected
  - Accident detected via relay.



Motor

Terminal 1 - Relay module (Common Contact)

Terminal 2 - Battery negative terminal

Battery

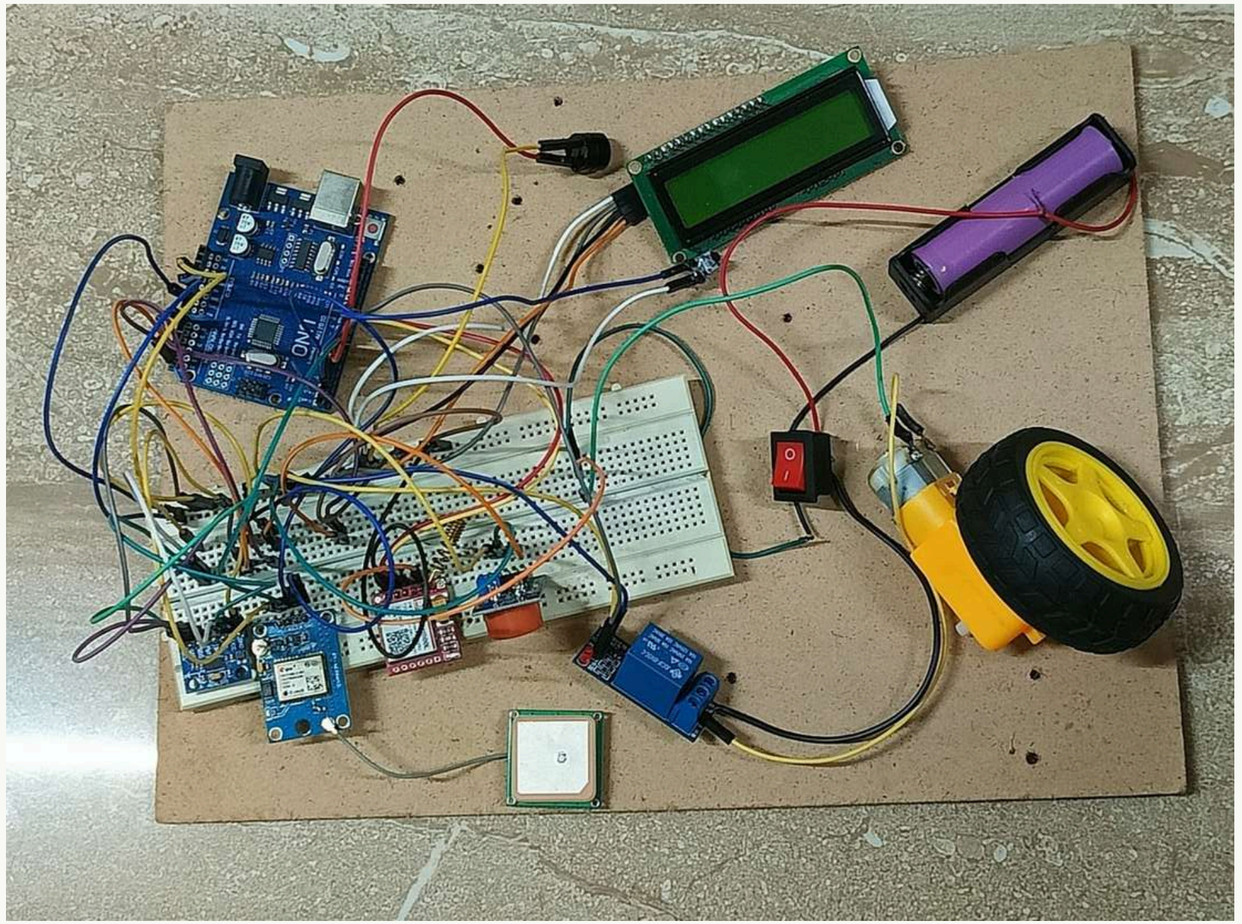
Battery positive terminal - GSM VCC and Switch Terminal 1

Relay Module NC - Switch Terminal 2

Working Principle :

1. System starts and reads alcohol sensor
2. If alcohol > threshold → engine lock
3. Otherwise accelerometer monitors motion
4. Crash detected → 10 second timer starts
5. If button pressed → alert cancelled
6. Otherwise GPS location is fetched
7. GSM sends emergency message

Image of the Project



Arduino Uno Code :

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_ADXL345_U.h>
#include <TinyGPS++.h>
#include <SoftwareSerial.h>
#include <math.h>
#define MQ3_A A0
#define RELAY 7
#define BUZZER 6
#define BUTTON 2
#define IMPACT_THRESHOLD 3
#define ALCOHOL_THRESHOLD 600
LiquidCrystal_I2C lcd(0x27,16,2);
Adafruit_ADXL345_Unified accel =
Adafruit_ADXL345_Unified(12345);
TinyGPSPlus gps;
SoftwareSerial gpsSerial(4,3);
SoftwareSerial gsm(9,8);
String phoneNumber = "+916303460839";
bool accidentDetected = false;
bool alertSent = false;
unsigned long accidentTime = 0;
void sendSMS(String message)
{
gsm.print("AT+CMGS=\"");
gsm.print(phoneNumber);
gsm.println("\"");
delay(1000);
gsm.print(message);
delay(500);
gsm.write(26);
delay(5000);
}
void makeCall()
{
gsm.print("ATD");
gsm.print(phoneNumber);
gsm.println(";");
delay(20000);
gsm.println("ATH");
}
void setup()
{
Serial.begin(9600);
gpsSerial.begin(9600);
gsm.begin(9600);
pinMode(MQ3_A,INPUT);
pinMode(RELAY,OUTPUT);
pinMode(BUZZER,OUTPUT);
pinMode(BUTTON,INPUT_PULLUP);
digitalWrite(RELAY,HIGH);
digitalWrite(BUZZER,LOW);
lcd.init();
lcd.backlight();
lcd.setCursor(0,0);
lcd.print("Smart Vehicle");
lcd.setCursor(0,1);
lcd.print("System Start");
delay(2000);
lcd.clear();
Serial.println("System Starting...");
// GSM initialization
gsm.println("AT");
delay(1000);
gsm.println("AT+CMGF=1");
delay(1000);
```

Arduino Uno Code :

```
if(!accel.begin())
{
Serial.println("ADXL345 ERROR");
lcd.setCursor(0,0);
lcd.print("ADXL Error!");
while(1);
}
accel.setRange(ADXL345_RANGE_16_G)
;
lcd.setCursor(0,0);
lcd.print("Sensor Warmup");
lcd.setCursor(0,1);
lcd.print("Please Wait");
delay(15000);
lcd.clear();
}
void loop()
{
// -------- GPS READ --------
while(gpsSerial.available())
gps.encode(gpsSerial.read());
// -------- ALCOHOL SENSOR --------
int alcoholLevel = analogRead(MQ3_A);
sensors_event_t event;
accel.getEvent(&event);
float x = event.acceleration.x;
float y = event.acceleration.y;
float z = event.acceleration.z;
float total = sqrt(x*x + y*y + z*z);
float impact = abs(total - 9.8);
// -------- SERIAL OUTPUT --------
Serial.print("Alcohol: ");
Serial.print(alcoholLevel);
Serial.print(" Impact: ");
Serial.println(impact);
// -------- LCD DISPLAY --------
lcd.setCursor(0,0);
lcd.print("A:");
lcd.print(alcoholLevel);
lcd.print(" I:");
lcd.print(impact,1);
lcd.print(" ");
// -------- ALCOHOL DETECTION --------
if(alcoholLevel >
ALCOHOL_THRESHOLD)
{
lcd.setCursor(0,1);
lcd.print("ENGINE LOCKED ");
digitalWrite(RELAY,LOW);
digitalWrite(BUZZER,HIGH);
delay(500);
return;
}
else
{
if(!accidentDetected)
{
digitalWrite(RELAY,HIGH);
digitalWrite(BUZZER,LOW);
}
}
// -------- ACCIDENT DETECTION --------
if(impact > IMPACT_THRESHOLD &&
!accidentDetected)
{
accidentDetected = true;
alertSent = false;
accidentTime = millis();
lcd.clear();
```

Arduino Uno Code :

```
lcd.setCursor(0,0);
lcd.print("ACCIDENT!");
lcd.setCursor(0,1);
lcd.print("Engine Stopped");
digitalWrite(RELAY,LOW);
digitalWrite(BUZZER,HIGH);
Serial.println("Accident
Detected");
}
// -------- ACCIDENT MODE --------
if(accidentDetected)
{
digitalWrite(RELAY,LOW);
digitalWrite(BUZZER,HIGH);
// CANCEL BUTTON
if(digitalRead(BUTTON) == LOW)
{
Serial.println("Accident
Cancelled");
lcd.clear();
lcd.setCursor(0,0);
lcd.print("Alert Cancelled");
digitalWrite(BUZZER,LOW);
accidentDetected = false;
alertSent = false;
delay(2000);
lcd.clear();
}
// -------- SEND ALERT --------
if((millis() - accidentTime > 10000) &&
!alertSent)
{
Serial.println("ALERT SENT!");
String message = "ACCIDENT ALERT!\n";
if(gps.location.isValid())
{
message +=
"Location:\nhttps://maps.google.com/?q=";
message += String(gps.location.lat(),6);
message += ",";
message += String(gps.location.lng(),6);
}
else
{
message += "GPS not available";
}
sendSMS(message);
makeCall();
lcd.clear();
lcd.setCursor(0,0);
lcd.print("Sending Alert");
lcd.setCursor(0,1);
lcd.print("Location Sent");
alertSent = true;
delay(3000);
lcd.clear();}}
// -------- GPS DISPLAY --------
if(!accidentDetected){
if(gps.location.isValid()){
lcd.setCursor(0,1);
lcd.print("Lat:");
lcd.print(gps.location.lat(),2);
delay(1500);
lcd.setCursor(0,1);
lcd.print("Lng:");
lcd.print(gps.location.lng(),2);
lcd.print(" ");}
else{
lcd.setCursor(0,1);
lcd.print("Waiting GPS... ");}}
delay(500);}
```

Advantages:

1. Automatic accident detection
2. Faster emergency response
3. Prevents drunk driving
4. Lowcost and reliable system

Conclusion:

1. This system improves road safety by combining alcohol detection

2.Accident detection and automatic emergency alert using GPS and GSM.

CURRENT IMPLEMENTATION STATUS

The project has been implemented as a working prototype using Arduino UNO.

Completed Modules

✓ Alcohol detection using MQ3 sensor

✓ Accident detection using ADXL345 accelerometer

✓ Engine locking using relay module

✓ Buzzer alert for accident and alcohol detection

✓ LCD display showing system status

✓ Push button for cancelling false accident alerts

Partially Tested Modules

✓ GPS module for location tracking

✓ GSM module for sending SMS alerts with location

Ideas for further development for the project

1. Driver Health Monitoring System

Integrates BP and pulse sensors to continuously monitor the driver’s vital signs.

The system compares readings with safe predefined health ranges.

2. Driver Dizziness / Fatigue Detection

Uses sensors or camera-based monitoring to detect driver dizziness, drowsiness, or fatigue.

Indicators such as head tilt, abnormal movement, or inactivity are analyzed.

3. Smart Geo-Based Speed Regulation

Uses GPS location tracking to identify sensitive zones such as:

School zones,Accident-prone areas,Hospital zones

The system automatically checks the allowed speed limit for that location.

Even if the driver attempts to accelerate beyond the limit:

The system restricts the engine speed.

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P.Rama TejaSri - Documentation

C.Haritha - Hardware circuit design and component connections

N.Swetha - Arduino programming and sensor interfacing

A.Charishma - GPS and GSM module integration and testing

Thank You